Question 1 Exercise 1 Page 1

1. SELECT Statement

```
SELECT *
FROM employees_db;
```

id	First_name	Last_name	department	Salary	hire_date	City
1.	John	Doe	IT	55000	2018-06-15	New York
2.	Jane	Smith	HR	48000	2019-07-20	Chicago
3.	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4.	Sarah	Brown	IT	53000	2021-03-25	New York
5.	David	White	Marketing ~~52000~~	52000	2016-04-10	San Francisco
6.	Emily	Davis	IT	62000	2015-02-14	Chicago
7.	Robert	Wilson	Finance	59000	2019-10-01	Houston
8.	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9.	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

2. SELECT DISTINCT STATEMENT

```
SELECT DISTINCT department
FROM employees_db;
```

Department
IT
HR
Finance
Marketing

3. ORDER BY Statement

```
SELECT firstname,
       last_name
FROM employees_db
~~ORDER~~ ORDER BY Salary DECS;
```

First_name	Last_name	Salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59 000
John	Doe	55000
Daniel	Clark	53000
Sara	Brown	53000
David	White	52000
James	Moore	51000
Laura	Hall	5000

4. LIMIT statement

```
SELECT*
FROM employees_db
ORDER BY Salary DESC
LIMIT 6;
```

Id	First_name	Last_name	department	Salary	hire_date	City
6	Emily	Davis	IT	62000	2015-02-14	Chicago
3	Nick	Johnson	Finance	60000	2017-09-20	New York
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2021-03-25	San Francisco

5. WHERE Statement

```
SELECT*
FROM Employees_db
WHERE depart='IT';
```

Id	first_name	Last_name	department	Salary	hire_date	City
1.	John	Doe	IT	55000	2018-06-15	New York
4.	Sarah	Brown	IT	53000	2021-03-25	New York
6	Emily	Davis	IT	62000	2015-02-10	San Francisco
10	Laura	Hall	IT	58000	2020-02-10	San Francisco

6. AND Statement

```
SELECT
    id,
    First_name,
    Last_name,
    department,
    Salary,
    hire_date,
    City
FROM employees_db
WHERE department = 'Finance' AND Salary > 58,000;
```

Id	First_name	Last_name	department	Salary ~~hire_date~~	hire_date ~~City~~	City
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
7	Robert	Wilson	Finance	59000	2019-10-01 ~~59000~~	Houston

7. OR Statement

```
SELECT*
FROM employees_db
WHERE department='HR' OR department='Marketing';
```

Id	First_name	Last_name	department	Salary	hire_date	City
2.	Jane	Smith	HR	48000	2019-07-20	Chicago
5.	David	White	Marketing	52000	2016-04-10	San Francisco
8	Jessica	Moore	HR	57000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago

8 Not Statement

```
SELECT *
FROM employees_db
WHERE NOT department='IT';
```

Id	first_name	Last_name	department	Salary	hire_date	City
2.	Jane	Smith	HR	48000	2019-07-20	Chicago
3.	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
5.	Daniel	White	Marketing	52000	2016-04-10	San Francisco
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	57000	2018-05-2022	Los Angeles
9.	Daniel	Clark	Marketing	53000	2022-06-01	Chicago

9. In Statement

```
SELECT *
FROM Employees_db
WHERE department IN ('HR', 'IT', 'Finance');
```

Id	First_name	Last_name	department	Salary	hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
6.	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	IT	50000	2020-08-10	San Francisco

10. Combining Conditions

```
SELECT *
FROM employees_db
WHERE department = 'IT'
      AND Salary > 50000
      AND City = 'New York';
```

Id	First_name	Last_name	department	Salary	hire_date	City
1.	John	Doe	IT	55000	2018-06-15	New York
4.	Brown	Brown	IT	53000	2021-03-25	New York

11. Combing WHERE, AND, and ORDER BY

```
SELECT first_name,
       Last_name
FROM employees_db
WHERE department = 'Finance' AND department = 'Marketing'
      ~~IN ('Finance', 'Marketing')~~
      AND Salary > 52000
ORDER BY Salary DESC;
```

first_name	last_name	department	Salary
Mike	Johnson	Finance	60000
Robert	Wilson	Finance	59000
David	Clark	Marketing	53000

12 Combining SELECT DISTINCT, WHERE and IN

```
SELECT DISTINCT City
FROM employees_db
WHERE department NOT IN ('IT', 'HR');
```

City
Los Angeles
San Francisco
Houston
Chicago

13. Combining WHERE, NOT, AND, and ORDER BY

```
SELECT *
FROM employees_db
WHERE NOT department = 'Finance'
      AND Salary > 50 000
ORDER BY hire_date ASC;
```

id	last_name	first_name	department	Salary	hire_date	City
6	Emily	Davis	IT	62000	2015-02-14	Chicago
5	David	White	Marketing	52000	2016-04-10	San Francisco
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2021-03-25	New York
9	Daniel	clark	Marketing	53000	2022-06-01	Chicago

14. Combining WHERE, OR, IN and LIMIT

```
SELECT *
FROM employees_db
WHERE City = 'Chicago' OR City = 'Los Angeles'
AND department = 'IT' OR department = 'Marketing'
LIMIT 3;
```

id	first_name	Last_name	department	Salary	hire_date	City
6	Emily	Davis	IT	62000	2015-02-14	Chicago
9	Daniel	Clark	Marketing	55000	2022-06-01	Chicago